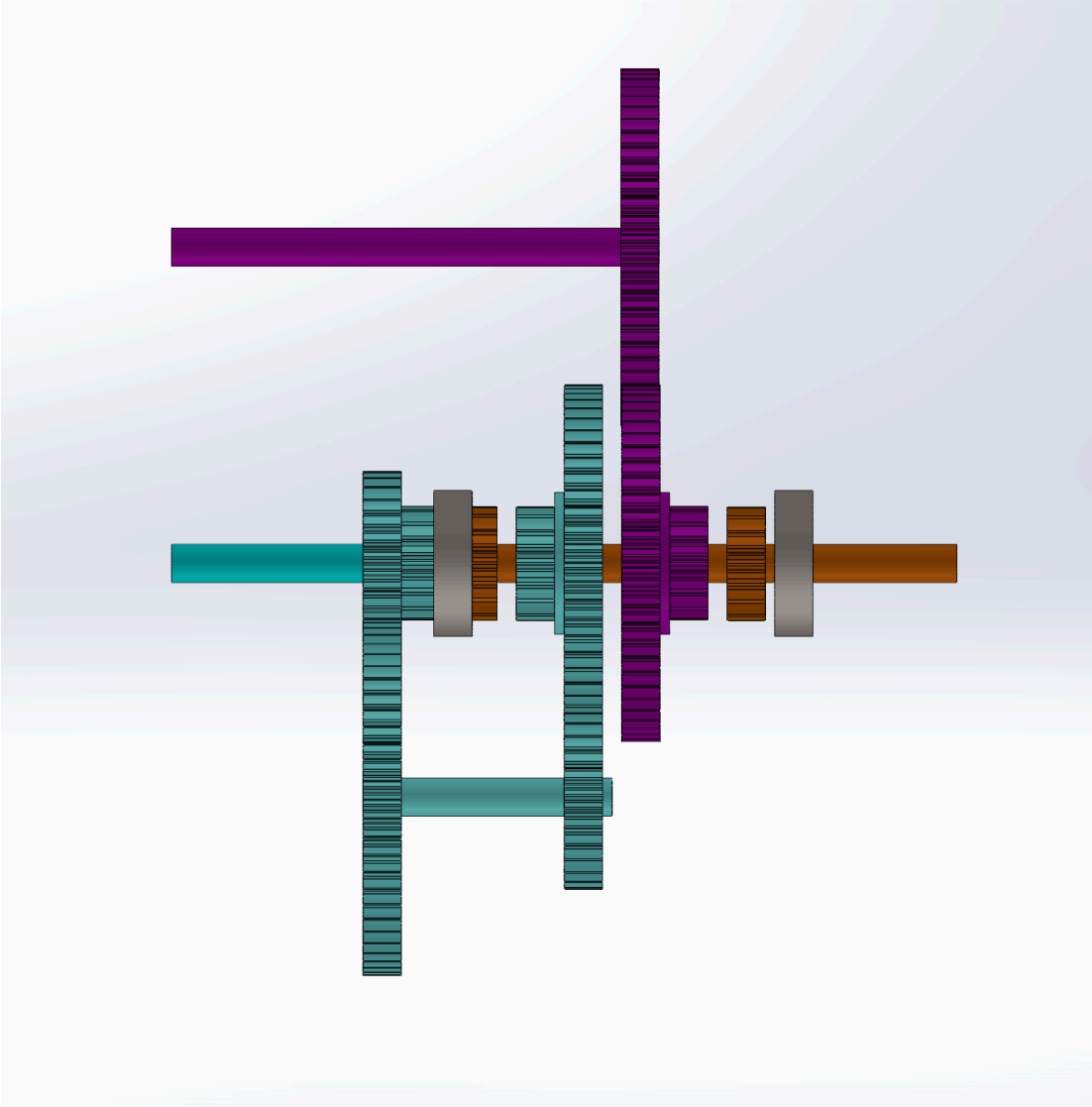
Phase 2
Transfer Case

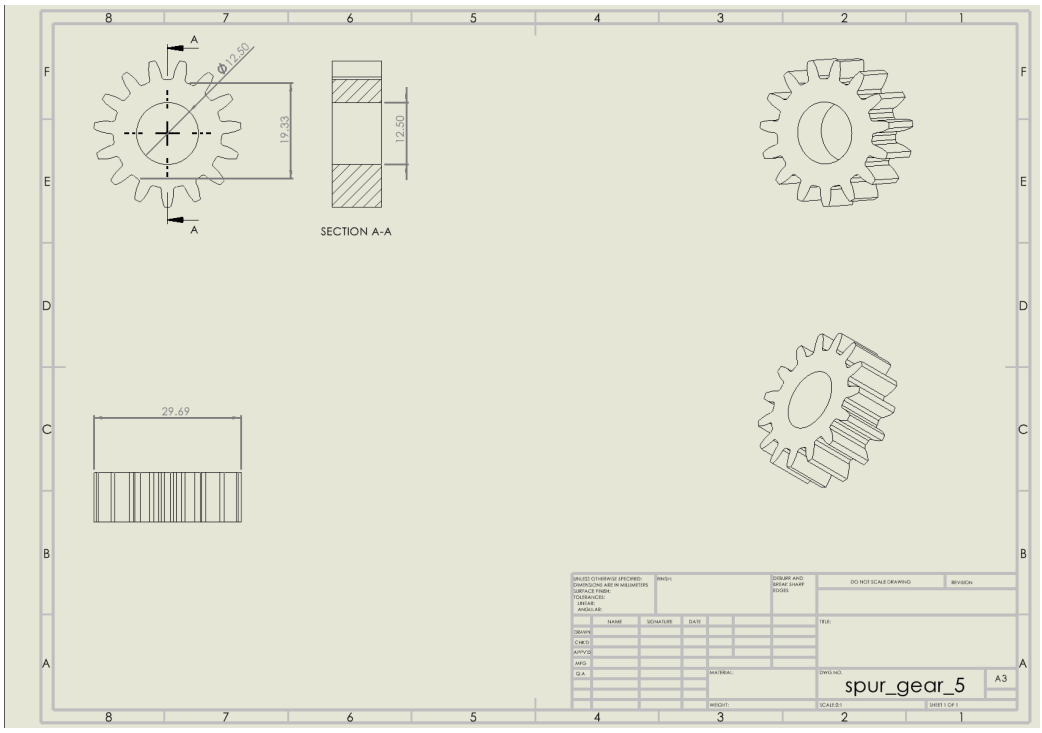
The transfer case is a crucial component in four wheel drive and all wheel drive vehicles that splits power from the transmission and sends it to both the front and rear axles, driveshafts. Its main function is to enable multiple axle power delivery, providing enhanced traction, and allowing drivers to switch between two wheel drive and four wheel drive, or high/low gear ranges.

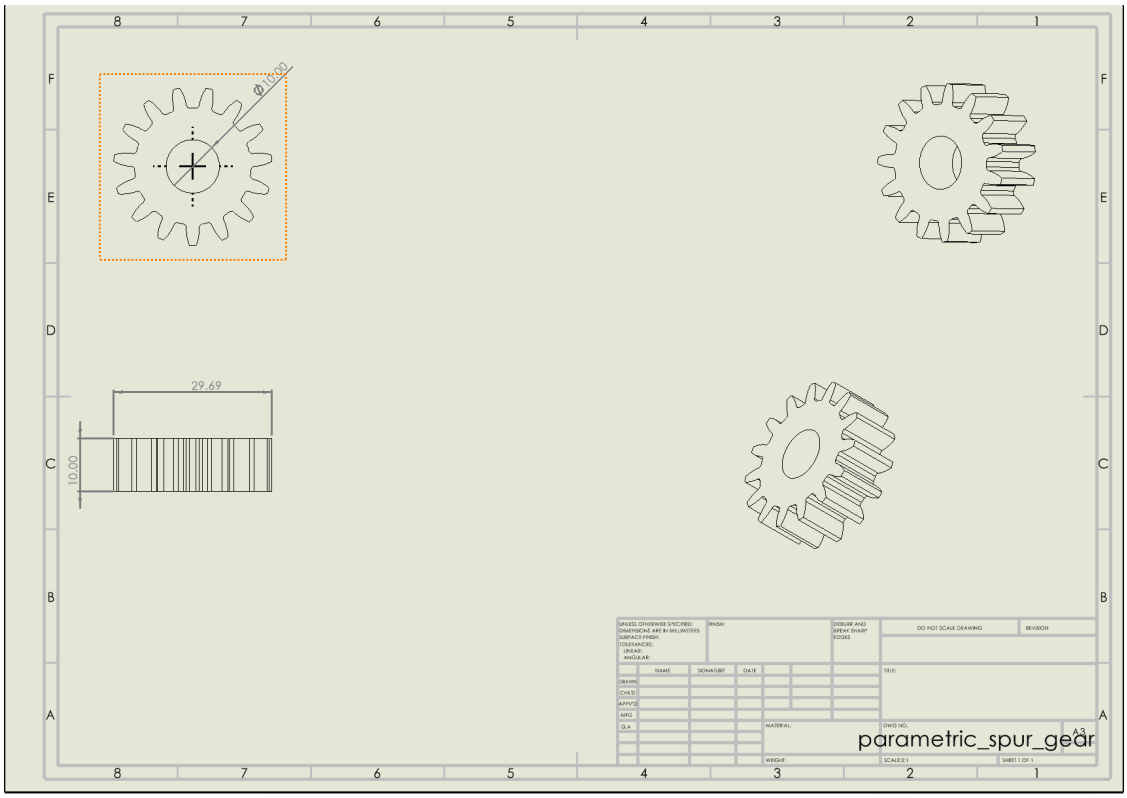
- **Full 3D assembly**
  - Insert image of the SOLIDWORKS assembly here



- **Drawings and views (SOLIDWORKS 2D)**
  - Insert Images of each part of the assembly with dimensions and tolerances.

- Make one exploded view of the assembly
- Make a basic motion study showing how it moves

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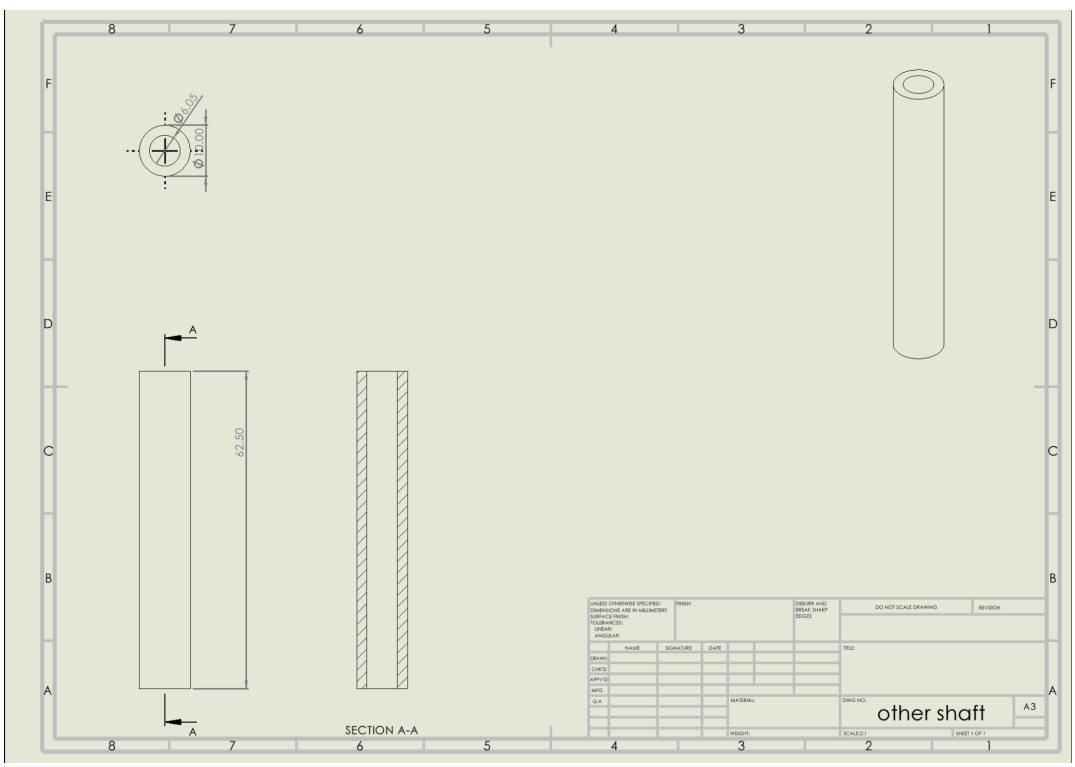
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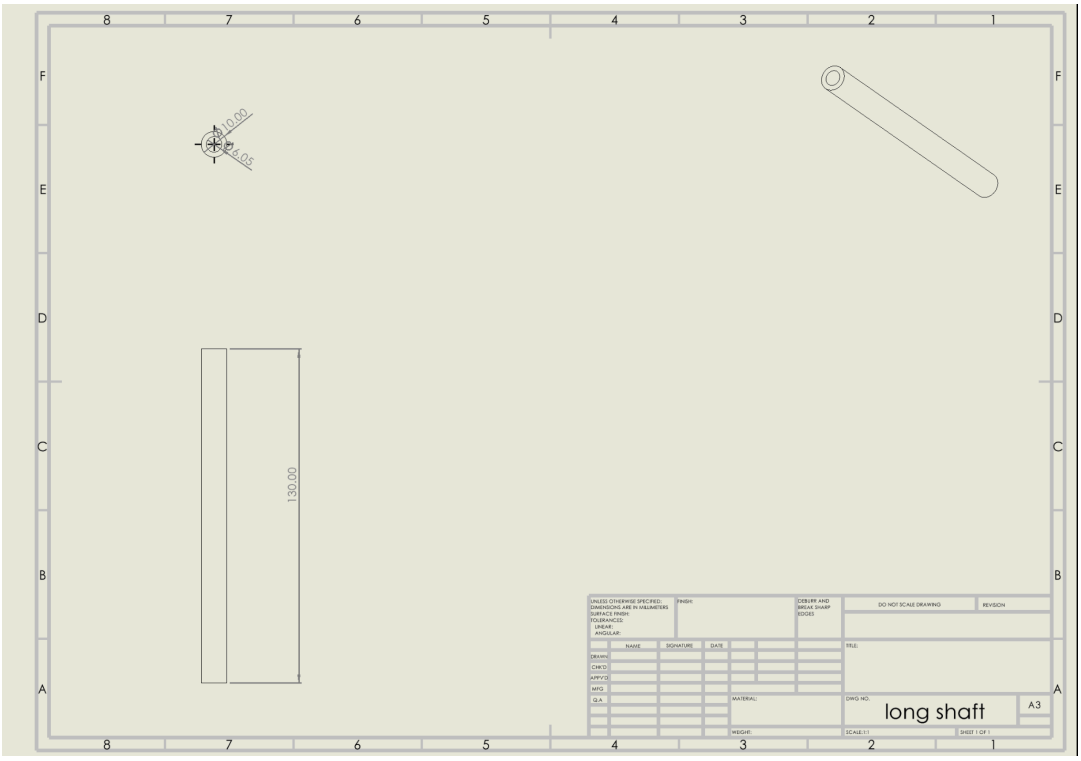
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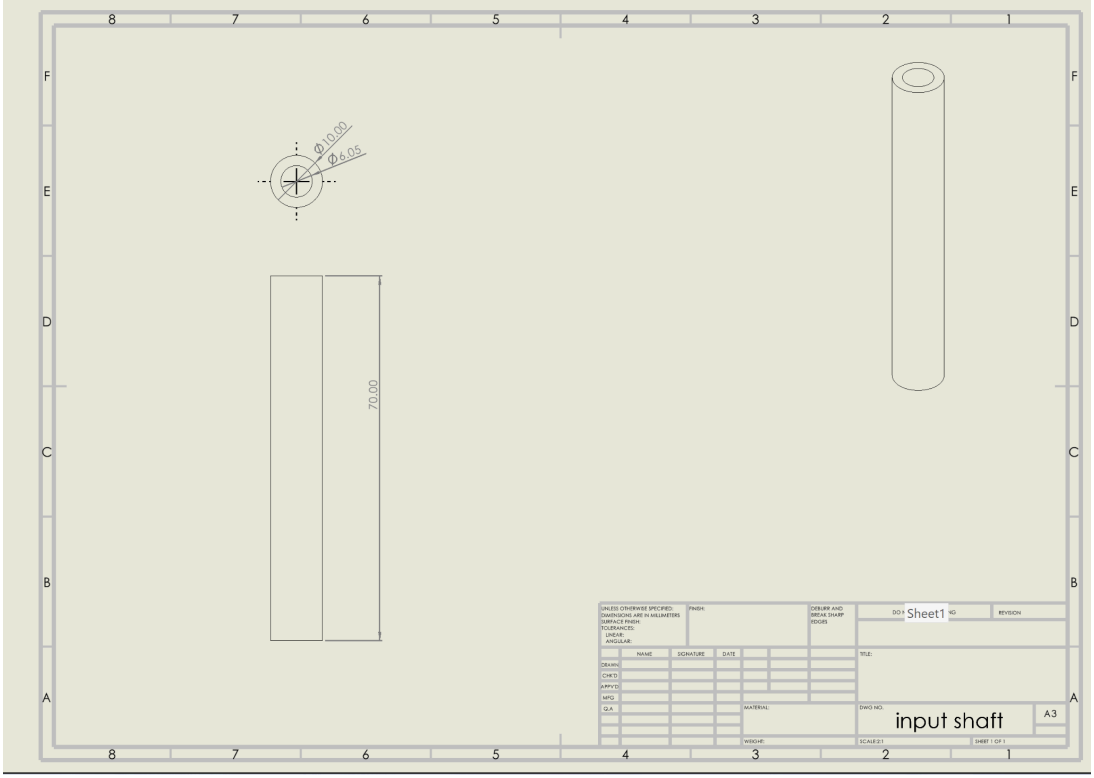
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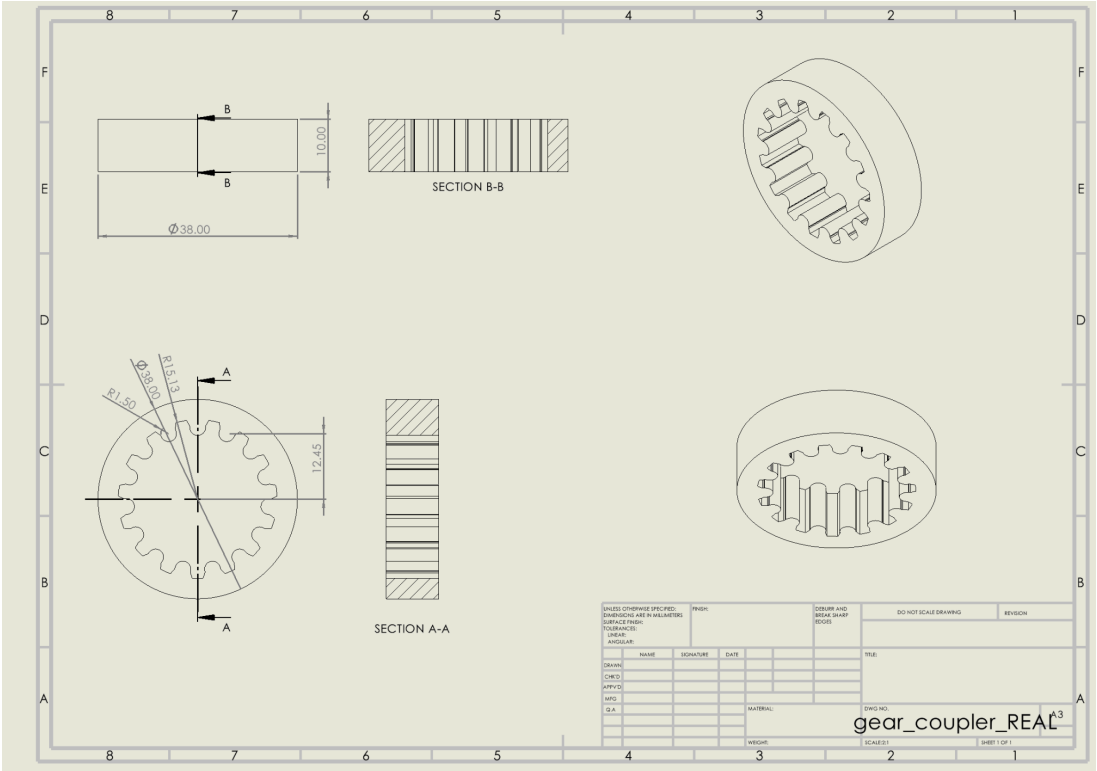
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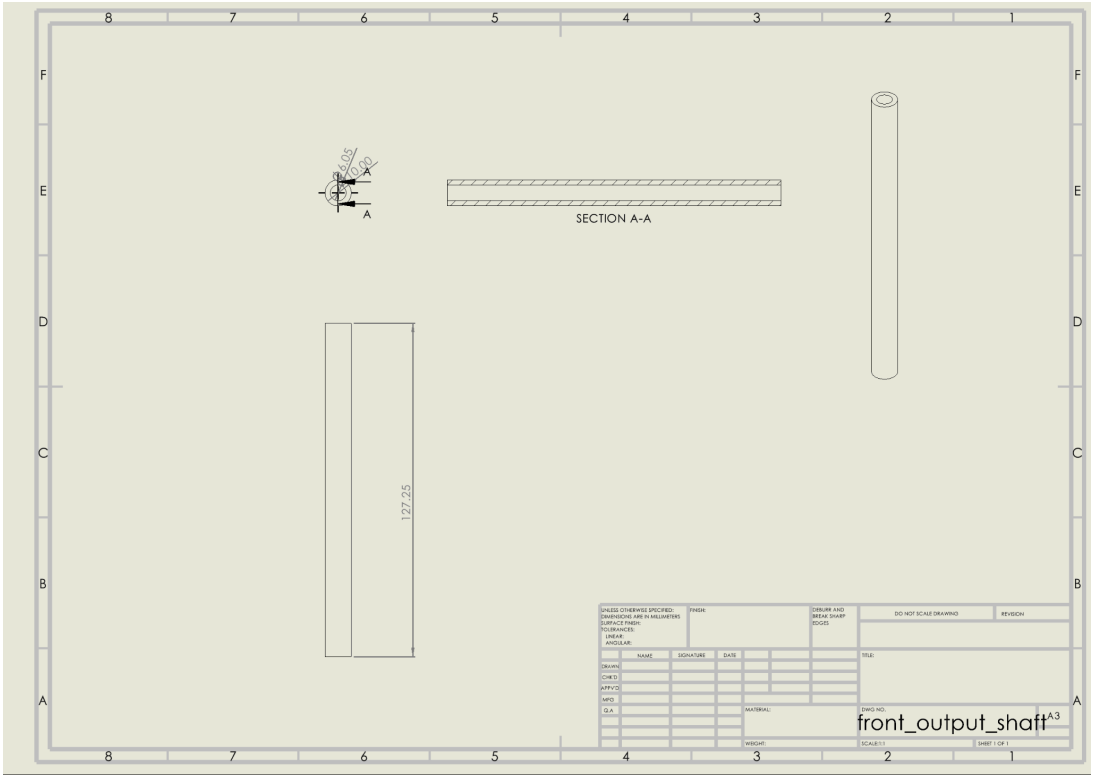
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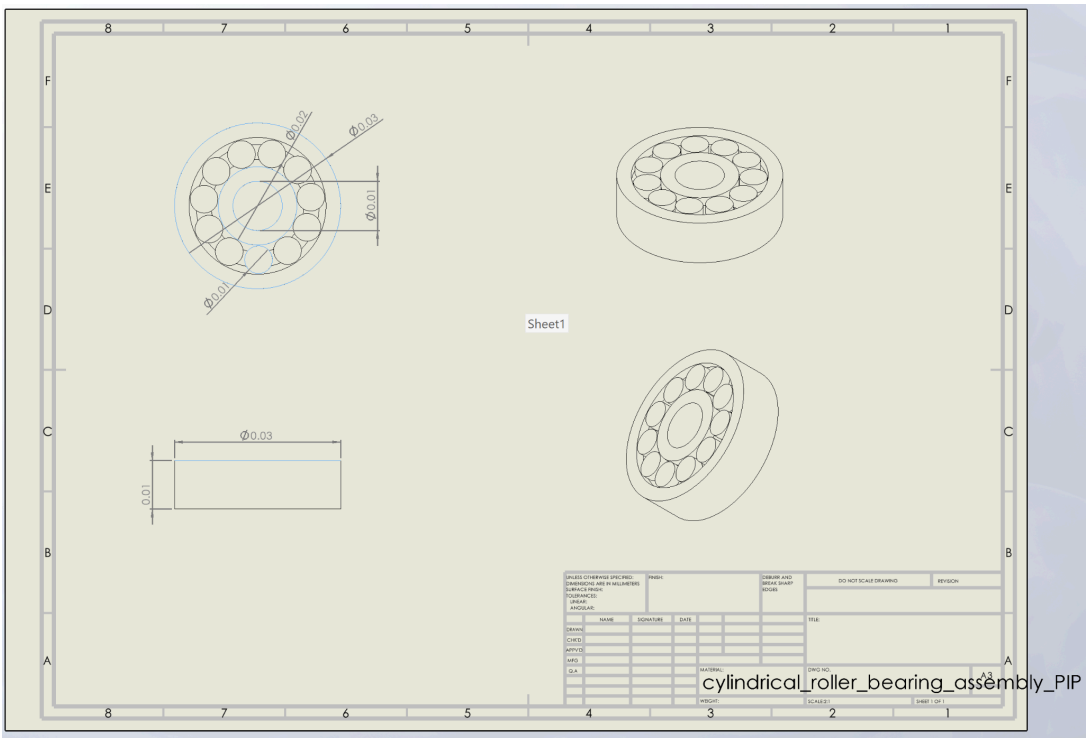
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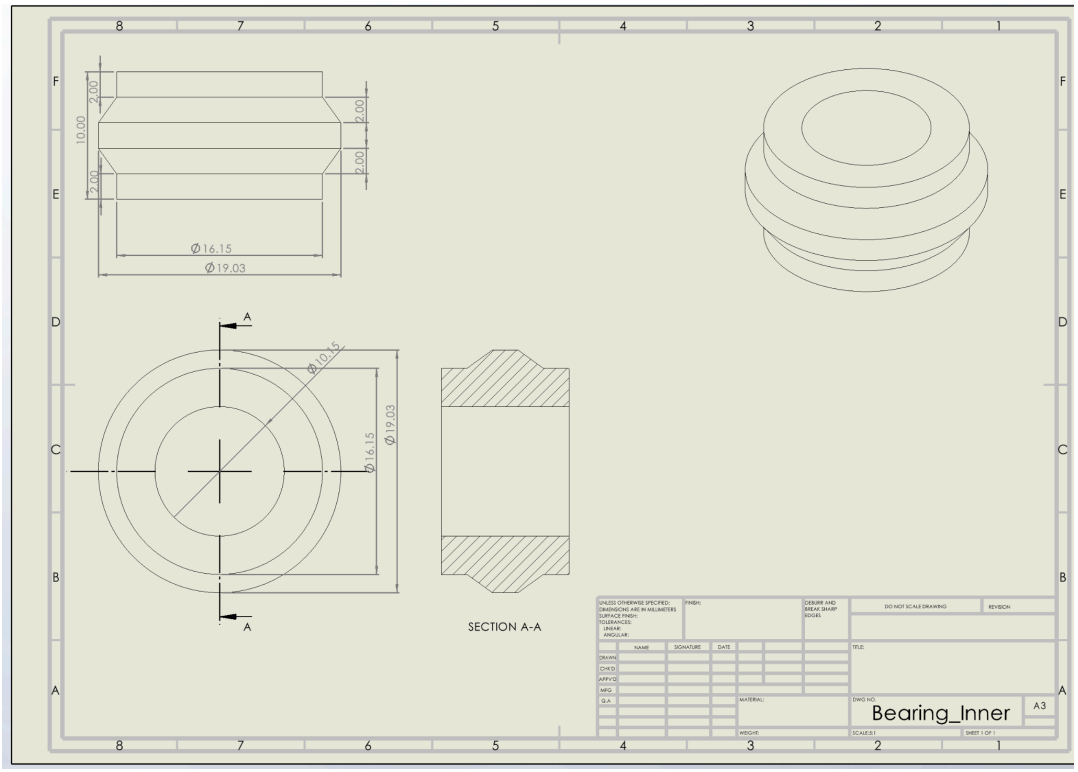
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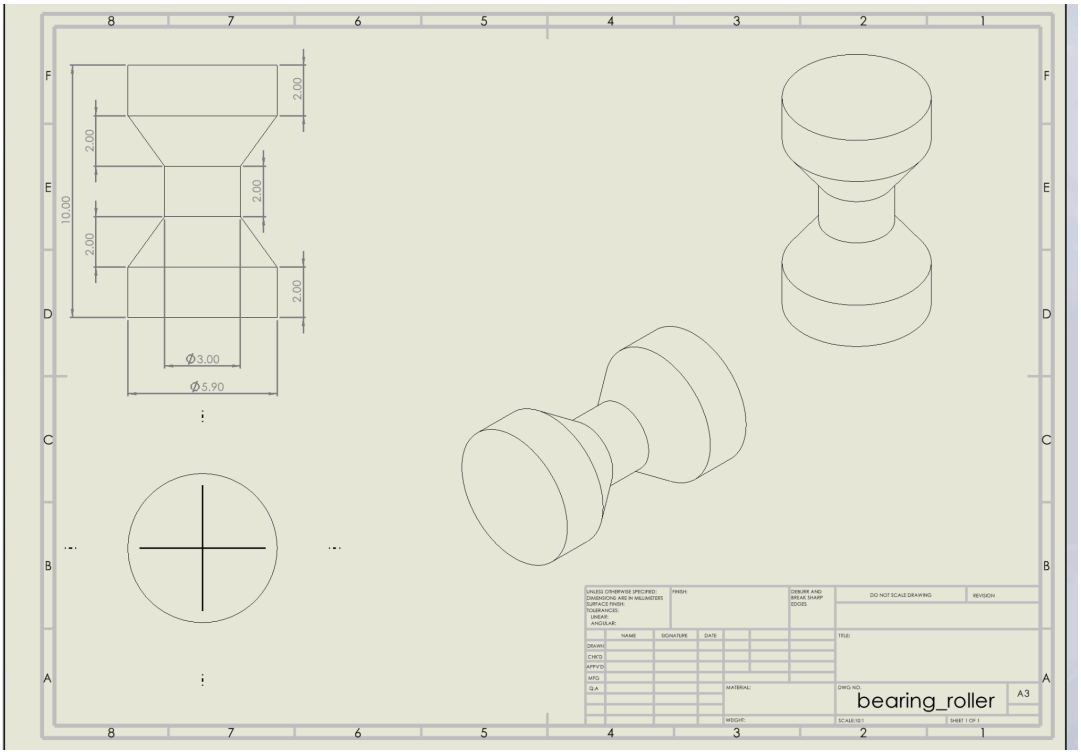
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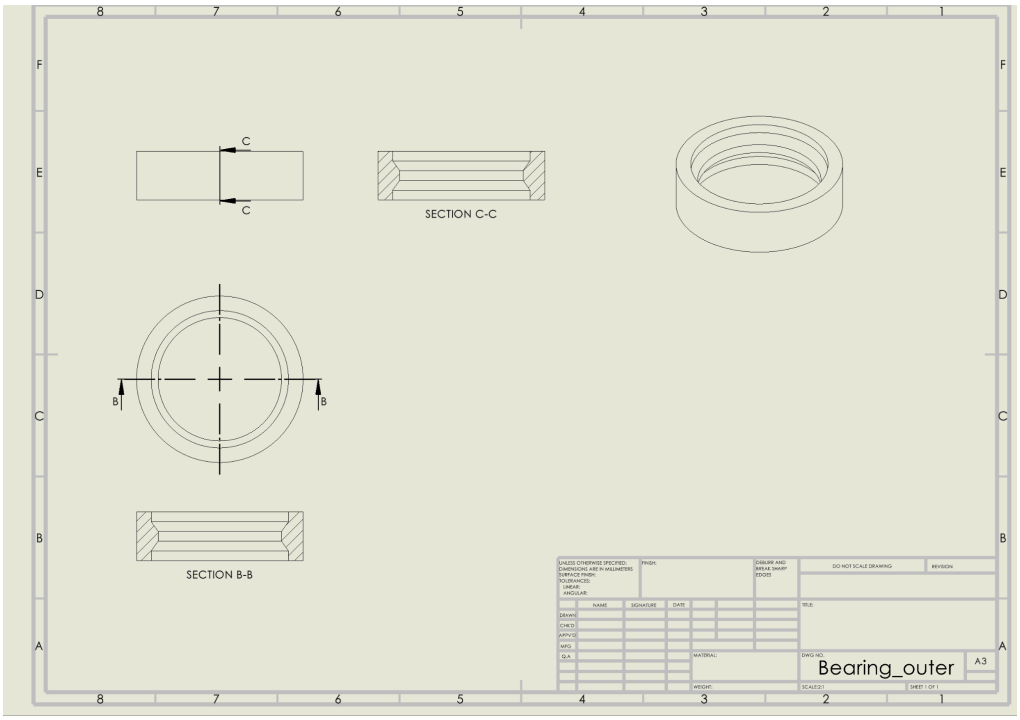
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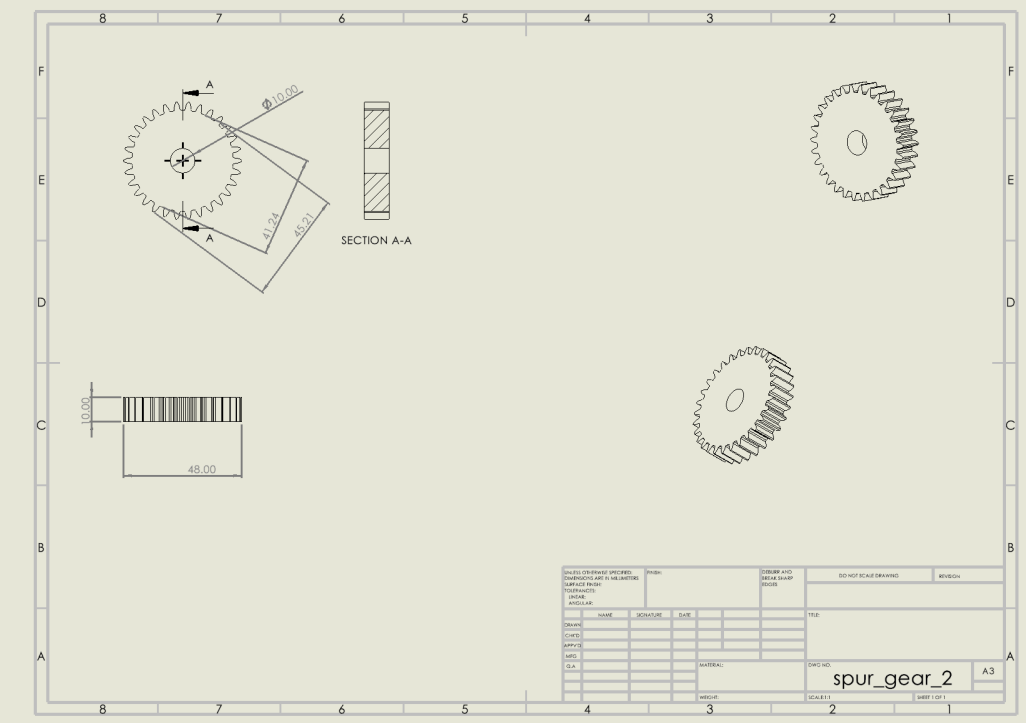
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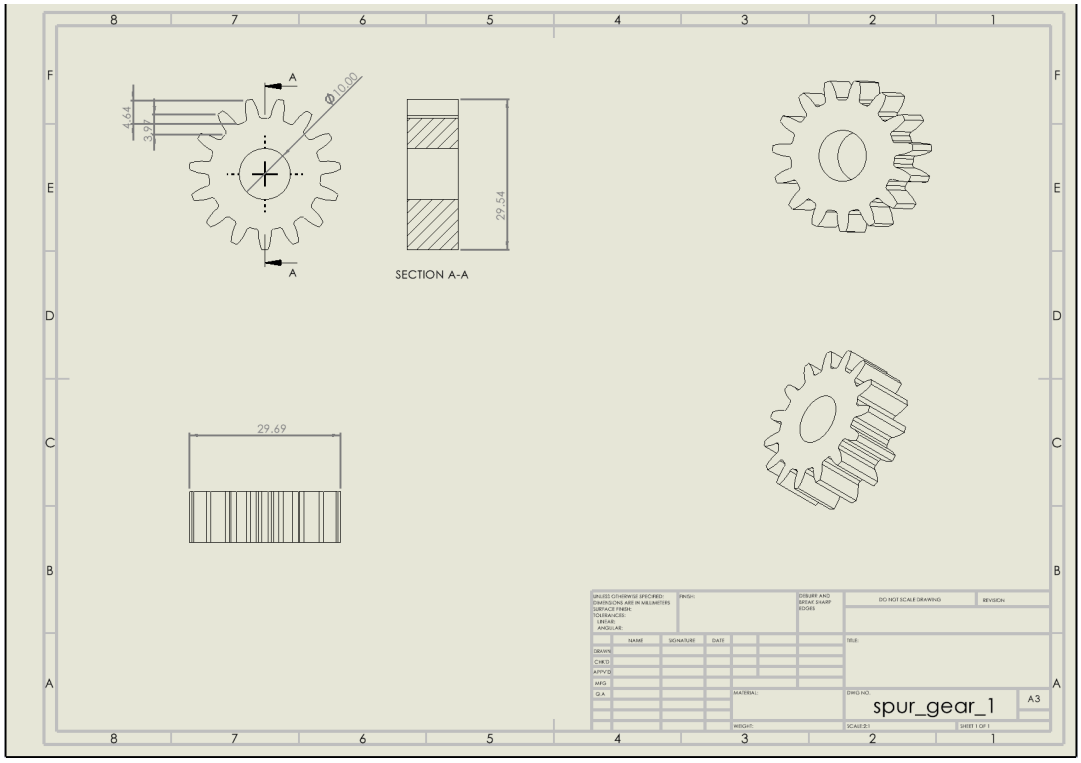
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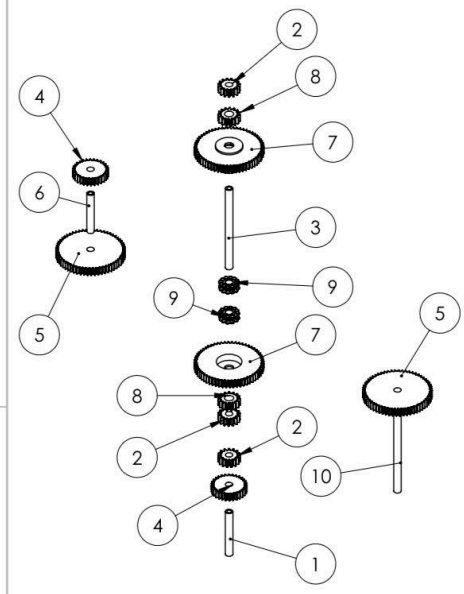

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Input shaft	4130 Steel	1
2	Spur Gear 1	4140 Steel	3
3	Long Shaft	4130 Steel	1
4	Spur Gear 2	4140 Steel	2
5	Spur Gear 3	4140 Steel	2
6	Medium shaft	4130 Steel	1
7	Spur Gear 4(W/bearing)	4140 Steel	2
8	Spur Gear 5	4140 Steel	2
9	Cylindrical Roller Bearing Assembly	301 Stainless Steel	2
10	Front Output Shaft	4130 Steel	1

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NEXT ASSY | USED ON
APPLICATION

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±
INTERPRET GEOMETRIC TOLERANCING PER:
MATERIAL
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CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		

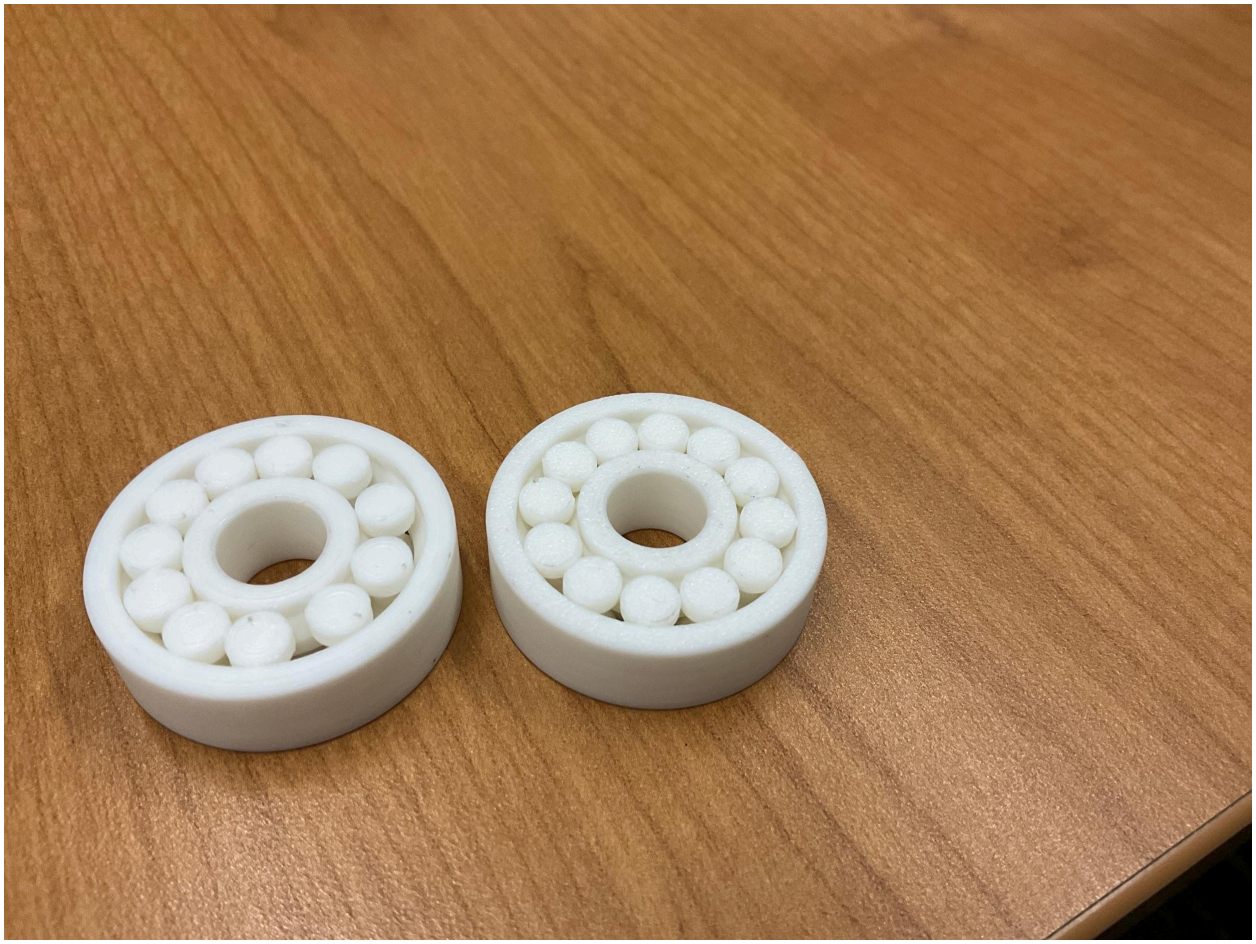
COMMENTS:

TITLE:

SIZE A | DWG. NO. Assem1 | REV
SCALE: 1:12 | WEIGHT: | SHEET 1 OF 1

SOLIDWORKS Educational Product. For Instructional Use Only.

- **Printability (Show ball bearing prints)**
  - Make sure all the parts are within the build constraints of a 3D Lab printer
  - Design with appropriate feature sizes and clearances for chosen filament and 3D printing resolution



The transfer case was designed with 3D printability in mind by ensuring that all components fit within the available printer build volume and by using practical feature sizes and clearances for the selected printing process. To validate the design, prototype ball bearings were 3D printed and inspected after fabrication. The print bearings were structurally solid and retained their intended geometry, indicating that the design is feasible for physical prototyping. This print test also provided confidence that the chosen tolerances and part dimensions are suitable for assembly and functional evaluation.

Ansys Analysis

- **static stress and factor of safety**
- **Fatigue assessment**
- **Gear tooth loading (if applicable**
    - Estimate tangential tooth forces from torque and pitch radius
    - Preform bending stress check for gear teeth
    - Discuss how contact stress and wear affect your design
- **Coupling stresses (if applicable)**
    - For at least one shaft or shaft gear connection, compute approximate bearing and
- **Bearing load check**

    - Determine reaction loads at one bearing set based on shaft loading and gear forces
    - Check against basic bearing capacity data and argue qualitatively how long the bearing would be expected to last at your reduced loads
- **Global safety overview**
    - Summarize critical locations, their computed factors of safety (static and fatigue), and any design modifications you made to achieve acceptable margins